

CPU Scheduling, Synchronization & Deadlocks - Quick Revision

QUICK REVISION & EXAM CHEAT SHEET

Subject: Operating Systems | Chapter: Chapter 2: CPU Scheduling, Deadlocks & Synchronization | Academic Year: 2026-2027

Scheduling Formulas

- * Turnaround Time (TAT) = Completion Time - Arrival Time.
- * Waiting Time (WT) = Turnaround Time - Burst Time.
- * Response Time = First Time CPU Allocated - Arrival Time.

Critical Section Requirements

- * 1. Mutual Exclusion: At most one process executes inside the critical section.
- * 2. Progress: If no process is in critical section, only processes wishing to enter decide who enters next.
- * 3. Bounded Waiting: Bound exists on number of times other processes can enter before a waiting process enters.

Deadlock Prevention & Banker Algorithm

- * Deadlock requires: Mutual Exclusion + Hold & Wait + No Preemption + Circular Wait.
- * Banker Algorithm: Calculates $\text{Need}[i][j] = \text{Max}[i][j] - \text{Allocation}[i][j]$. Safe state if all processes can finish.